

HYPOTHESIS TESTING

HYPOTHESIS TESTING ON THE POPULATION MEAN (μ) of a CONTINUOUS VARIABLE.

Example: George, a well-known chicken farmer, has invented a new chicken food supplement. He claims it increases their weight. Previously the mean weight of his chicken population was 2.5 kg with standard deviation 0.4 kg. A sample of 60 chickens fed the supplement had a mean weight of 2.56 kg. Should George's claim be accepted at the 5% significance level?

Terms and definitions

- H_0 The Null Hypothesis = 'no change'
- H_1 The Alternative Hypothesis = 'change' (an increase, decrease or either)

If one is true then the other is false.

Basic principle: is what we have observed 'extreme enough' for us to decide that H_0 is false

- The test statistic is calculated from the sample. Its value is used to decide whether the null hypothesis, H_0 , should be rejected. $\bar{x} = 2.56 \text{ kg}$
- The rejection (or critical) region gives the values of the test statistic for which the null hypothesis, H_0 , is rejected.
- The acceptance region gives the values of the test statistic for which the null hypothesis, H_0 , is not rejected.

- The boundary value(s) of the rejection region is (are) called the critical value(s).

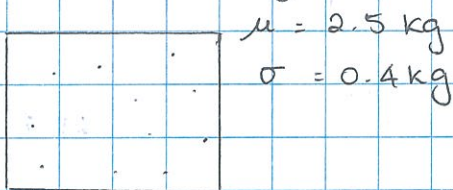
5%, 10%, 1%.

- The significance level of a test gives the probability of the test statistic falling in the rejection region when H_0 is true.

To carry out a hypothesis test:

- Step 1 Define the null and alternative hypotheses
- Step 2 Decide on a significance level.
- Step 3 Determine the critical value(s)
- Step 4 Calculate the test statistic
- Step 5 Decide on the outcome of the test depending on whether the value of the test statistic is in the rejection or the acceptance region.
- Step 6 State the conclusion in words

Method 1 : Comparing observed sample mean with critical regions



Step 1 $H_0 : \mu = 2.5 \text{ kg}$

$H_1 : \mu > 2.5 \text{ kg}$

2 Sig. level $\alpha = 5\%$

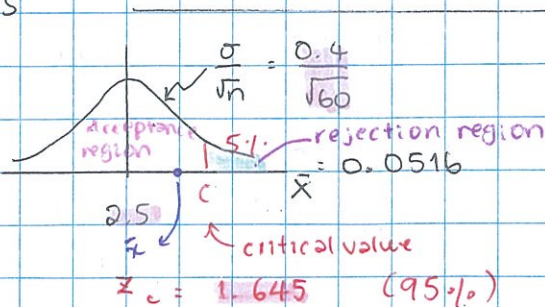
3 Critical values

If H_0 is true :

Distribution of sample means

$$Z = \frac{C - \mu}{\left(\frac{\sigma}{\sqrt{n}}\right)}$$

$$1.645 = \frac{C - 2.5}{\left(\frac{0.4}{\sqrt{60}}\right)}$$



Step 4 : test statistic $\bar{x} = 2.56 \text{ kg}$.

5. compare $\bar{x}$ with C

$2.56 < 2.585$ $\bar{x}$ is not in rejection region

SO, we accept H_0 (no change)

6.

The chicken supplement does not work

$$\Rightarrow C = 2.58494 \dots = 2.585$$

Observations over a long period of time have shown that the mass of adult males of a type of bat is normally distributed with mean 110g and standard deviation 10g. A scientist has a theory that in one area these bats are becoming smaller, possibly as an adaption to changes in their environment. He plans to trap 20 adult male bats, weigh them and then release them. He will then use the data to carry out a suitable hypothesis test at the 5% sig. level.

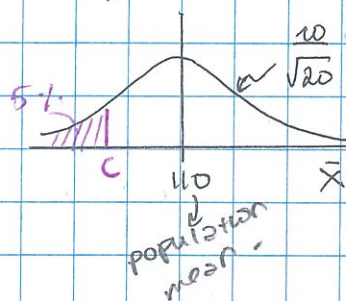
(i) State the null and alternative hypotheses.

(ii) Find the critical value for the test.

$$(1) H_0: \mu = 110 \text{ g}$$

$$H_1: \mu < 110 \text{ g}$$

$$\alpha = 5\%$$



$$Z = \frac{c - \mu}{\frac{\sigma}{\sqrt{n}}}$$

$$-1.645 = \frac{c - 110}{\frac{10}{\sqrt{20}}}$$

$$c = 106.3$$

$$\mu = 110 \quad \sigma = 10 \quad n = 20$$

$$\bar{x} = 106.3$$

the chicken farmer

question

Method 2 convert observed sample mean to a Z-value and compare with the critical Z-value.

Observed sample mean = 2.56

convert using $Z = \frac{\bar{x} - \mu}{\left(\frac{\sigma}{\sqrt{n}}\right)}$

$$= \frac{2.56 - 2.5}{\frac{0.4}{\sqrt{60}}}$$

$$= 1.162$$

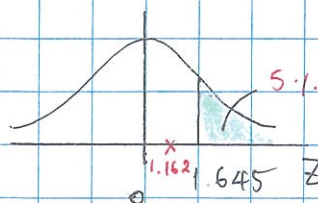
$$Z_c = \text{critical Z value for } 5\% \text{ test} = 1.645$$

$$\text{observe } Z = 1.162$$

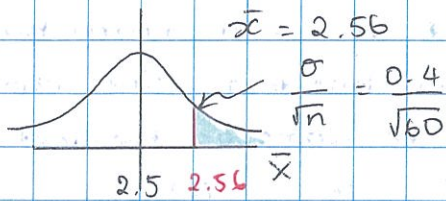
$$1.162 < 1.645$$

So we accept H_0 ,

$\therefore$ The chicken supplement does not work



Method 3 calculating probability of $\bar{x} \geq$ observed value and comparing with significance level.



$$\begin{aligned}
 P(\bar{x} > 2.56) \\
 &= P(Z > \frac{2.56 - 2.5}{\frac{0.4}{\sqrt{60}}}) \\
 &= P(Z = 1.162) \\
 &= 1 - 0.8774 \\
 &= 0.123 \quad (12.3\%)
 \end{aligned}$$

since 12.3% $\neq$ 5%, we accept H_0

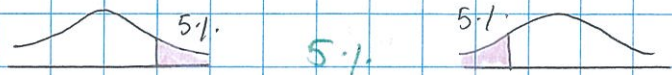
ONE-TAILED and TWO-TAILED TESTS

One-tailed tests are used when we wish to test whether there has specifically been an increase (or specifically a decrease) in the population mean.

$$H_0 = \mu$$

Keywords to look for: increase, decrease, reduction, lessening, lowering, raising...

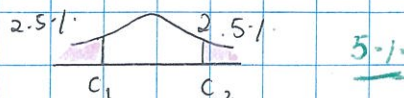
H_1 will be written as: $H_1: \mu > \text{number}$ or $H_1: \mu < \text{number}$



Two-tailed tests are used when we wish to test whether there has just been a change in the population mean.

Keywords: change, differ.

H_1 will be written as: $H_1: \mu \neq \text{number}$



A test of mental ability has been constructed so that, for adults in Great Britain, the test score is normally distributed with mean 100 and standard deviation 15. A doctor wishes to test whether sufferers from a particular disease differ in mean from the general population in their performance on this test. She chooses a random sample of 10 sufferers. Their scores on the test are

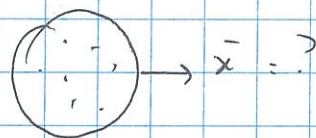
119 131 95 107 125 90 123 89 103 103

Carry out the test at the 5% significance level to test whether sufferers from the disease differ from the general population in the way in which they perform at this test.

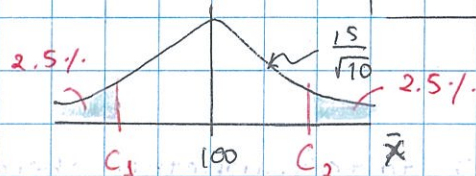
$$H_0: \mu = 100$$

$$H_1: \mu \neq 100$$

$\mu = 100$
 $\sigma = 15$



$$\bar{x} = 108.5$$



NOTE: calculate $\bar{x}$ first to see if C_1 or C_2 is needed to be calculated

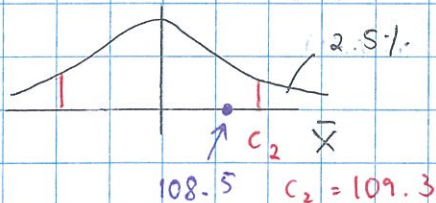
$z_c = 1.960$ the question is only interested in the top 2.5% because $\bar{x} = 108.5$

$$z_c = \frac{C_2 - \mu}{\left(\frac{\sigma}{\sqrt{n}} \right)}$$

$$1.960 = \frac{C_2 - 100}{\left(\frac{15}{\sqrt{10}} \right)}$$

$$C_2 = 109.3$$

since $108.5 < 109.3$ we accept H_0



Conclusion: these patients have a mental ability that is no different from the population

Example:

State the suitable null and alternative hypotheses

one tail $<, >$

two tail $\neq$

1. Bars of choco are claimed by the manufacturer to have a mean mass of 102.5 g. A test is carried out to see whether the mean mass of Choco bars is less than 102.5 g

$$H_0: \mu = 102.5 \quad H_1: \mu < 102.5$$

2. The mean factory assembly time for a particular electronic component is 84 s. It is required to test whether the introduction of a new procedure results in a different assembly time.

$$H_0: \mu = 84 \quad H_1: \mu \neq 84$$

3. In a report it was stated that the average age of all hospital patients was 53 years. A newspaper believes that this figure is an underestimation

$$H_0: \mu = 53 \quad H_1: \mu > 53$$

too small $\leftarrow$

4. The manufacturer of a certain battery claims that it has a mean life of 30 hours. A suspicious customer wishes to test the claim.

$$H_0: \mu = 30 \quad H_1: \mu < 30$$

5. A large batch of capacitors is judged to be satisfactory by an electronics factory if the mean capacitance is at least 5 microfarads. A test is carried out on a batch to determine whether it is satisfactory

$$H_0: \mu = 5 \quad H_1: \mu < 5$$

compare and with sig. level.

sig. level = 5%

$< 5\% \rightarrow$ reject H_0

$> 5\% \rightarrow$ accept H_0



HYPOTHESIS TESTING on DISCRETE VARIABLES

(i.e. Binomial or Poisson) countable number

probabilities are individuals rather than area

To carry out a hypothesis test on a discrete variable, calculate the probability of the observed or a more extreme value and compare this probability with the significance level. For a one tail test, reject the null Hypothesis if this probability is less than the significance level; for a two-tail test reject the null hypothesis if this probability is less than half of the significance level.

$$X \sim B(n, p)$$

$$P(X=a) = {}^nC_a p^a q^{n-a}$$

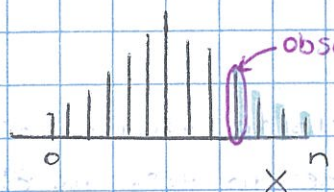
Binomial Variable:

$$H_0 : p = \square$$

$$H_1 : p < \square, p > \square, p \neq \square$$

one tailed

two-tailed



observed value

$$P(X \geq \text{observed value})$$

compare with sig. level

using Binomial formula

depends on H_1

Example: one tailed

A national opinion poll claims that 40% of the electorate would vote for party R if there were an election tomorrow. A student at a large college suspects that the proportion of young people who would vote for them is lower. She asks 16 fellow students, chosen at random from the college roll, which party they would vote for. Three choose party R. Test the student's suspicions at the 10% sig. level.

$$H_0 : p = 0.4$$

$$H_1 : p < 0.4$$

If H_0 is true, $X \sim B(16, 0.4)$

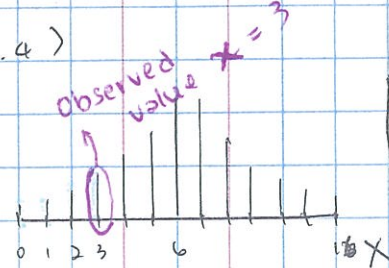
X = number voting party R



$$p = 0.4$$



$$n = 16$$



observed value $X=3$

$$\left[\begin{array}{l} \text{biggest bar} = np \\ = 16 \times 0.4 \\ = 6.4 \end{array} \right]$$

calculate $P(X \leq 3) = P(X = 0, 1, 2, 3)$

$$= {}^{16}C_0 (0.4)^0 (0.6)^{16} + {}^{16}C_1 (0.4)^1 (0.6)^{15} + {}^{16}C_2 (0.4)^2 (0.6)^{14} + {}^{16}C_3 (0.4)^3 (0.6)^{13}$$

$$= 0.00028... + 0.0030... + 0.01504... + 0.04680...$$

$$= 0.06514 = 6.51\%$$

compare 6.51% with sig level 10% .

Since $6.51\% < 10\%$, we reject H_0 .

So, conclude that proportion of young people voting for party R is less than $0.4 \rightarrow$ reject H_0 means accept $H_1: p < 0.4$

means H_1 is true.
10%
rejection area.

Example. 2-tailed.

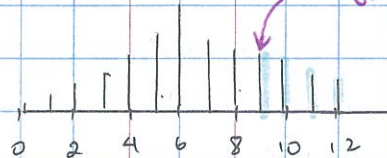
In order to test a coin for bias it is tossed 12 times. The result is 9 heads and 3 tails. Test at 10% significance level. Whether the coin is biased.

$$H_0: p = 0.5$$

$$H_1: p \neq 0.5$$

$$X = \text{no of heads}$$

$$X \sim B(12, 0.5)$$



$$np = 12 \times 0.5 = 6$$

observed value $X = 9$

if observed value $X = 2$ $P(X \leq 2)$

$$P(X \geq 9) = P(X = 9, 10, 11, 12)$$

$$= {}^{12}C_9 (0.5)^9 (0.5)^3 + {}^{12}C_{10} (0.5)^{10} (0.5)^2 + {}^{12}C_{11} (0.5)^{11} (0.5) + {}^{12}C_{12} (0.5)^{12} (0.5)^0$$

$$= 0.05371 + 0.01611 + 0.00293 + 0.00024$$

$$= 0.0729 = 7.3\%$$

compare 7.3% with half of sig. level $(\frac{1}{2} \text{ of } 10\%) = 5\%$.

Since $7.3\% > 5\%$, we accept H_0 .

The coin is not biased.

Isaac claims that 30% of cars in his town are red. His friend Hardip thinks that the proportion is less than 30%. The boys decided to test Isaac's claim at the 5% sig. level and found that 2 cars out of the random sample of 18 were red. Carry out the hypothesis test and state your conclusion.

9709, Paper 7, Q. Nov 2007.

$$\frac{2}{18} = 11\%$$

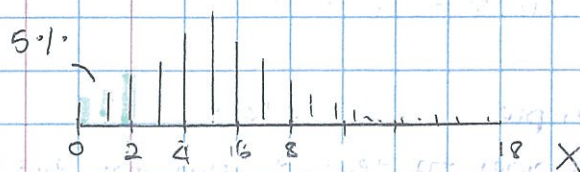
$$H_0 : p = 0.3$$

$$H_1 : p < 0.3$$

X = No. of the red car

$$X \sim B(18, 0.3)$$

$$np = 18 \times 0.3 = 5.4$$



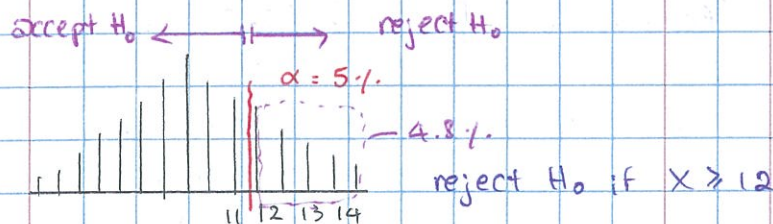
Observed $X = 2$.

$$\begin{aligned} P(X \leq 2) &= P(X = 0, 1, 2) \\ &= {}^{18}C_0 (0.3)^0 (0.7)^{18} + {}^{18}C_1 (0.3)^1 (0.7)^{17} + {}^{18}C_2 (0.3)^2 (0.7)^{16} \\ &= 0.001628... + 0.012562... + 0.04576... \\ &= 0.059949... = 0.06 \quad 6\% \end{aligned}$$

6% > 5%, so we accept H_0 .

So, 30% of cars in the town are red.

FINDING THE REJECTION REGION FOR A BINOMIAL TEST



Example: 1.

World-wide 25% of men are colour blind but it is believed that the condition is less widespread among a group of remote hill tribes. An anthropologist plans to test this by sending field workers to visit villages in that area. In each village 30 men are to be tested for colour-blindness. Find the rejection region for the test at 5% level of significance.

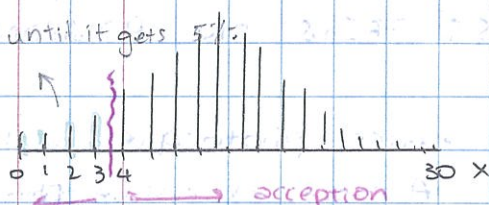
$$H_0 : p = 0.25$$

$$H_1 : p < 0.25$$

$X = \text{no of colour-blind}$

$$X \sim B(30, 0.25)$$

$$np = 30 \times 0.25 = 7.5$$



$$\begin{aligned} P(X=0) &= {}^{30}C_0 (0.25)^0 (0.75)^{30} = 0.0001785 \dots \\ P(X=1) &= {}^{30}C_1 (0.25)^1 (0.75)^{29} = 0.00176 \dots \\ P(X=2) &= {}^{30}C_2 (0.25)^2 (0.75)^{28} = 0.00863 \dots \\ P(X=3) &= {}^{30}C_3 (0.25)^3 (0.75)^{27} = 0.0269 \dots \\ P(X=4) &= {}^{30}C_4 (0.25)^4 (0.75)^{26} = 0.06042 \dots \end{aligned}$$

$$\begin{aligned} P(X \leq 3) &= 0.0375 \\ &= 3.8\% \end{aligned}$$

$\rightarrow P(X=4)$ itself $> 5\%$

So, rejection region : Reject H_0 if $x \leq 3$

Example 2 : In a certain country, 90% of letters are delivered the day after posting. It is later suspect that the service has deteriorated as a result of mechanisation. To test this, 17 letters are posted and it is found that only 13 of them arrived the next day.

(i) write the suitable null and alternative hypotheses.

$$H_0 : p = 0.9$$

$$H_1 : p < 0.9$$

$X = \text{no of letters deliver the next day}$

$$X \sim B(17, 0.9) \quad np = 15.3$$

(iii) write down the critical region for the test, giving reason for your choice.

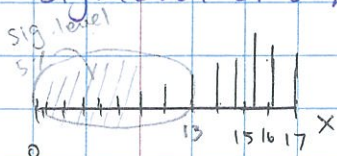
$$x \leq 13 (2.3\% < 5\%)$$

$P(X=13)$ itself $> 5\%$

$$\begin{aligned} P(X=13) &= {}^{17}C_{13} (0.9)^{13} (0.1)^4 \\ &= 0.06049 \dots \quad 6\% \end{aligned}$$

$$\begin{aligned} P(X \geq 13) &= 0.0604 \dots + 0.9173 \dots \\ &= 0.9777 \dots \quad 98\% \end{aligned}$$

(ii) carry out the hypothesis test at the sig. level of 5%.



observed $x = 13$

$$\begin{aligned} P(X \leq 13) &= 1 - P(X=14, 15, 16, 17) \\ &= 1 - (0.1556 \dots + 0.280 \dots + 0.3150 \dots + 0.1668 \dots) \\ &= 1 - 0.9173 \dots \\ &= 0.0826 \dots \quad 8.3\% \quad (8.3\% > 5\%) \end{aligned}$$

we accept H_0 , so 90% of the letters are delivered the next day after posting

$$\begin{aligned} \text{So, } P(X \leq 12) &= 1 - 0.978 \dots \\ &= 0.022 \dots \end{aligned}$$

$$2.2\% < 5\%$$

So, rejection region is $x \leq 12$

HYPOTHESIS TESTS ON POISSON VARIABLES

EX: The office photocopier fails on average three times every 2 weeks. A more expensive photocopier is claimed to be more reliable. In the first 4 weeks, the new copier fails once. Assuming poisson, test a 5% whether the new copier is more reliable

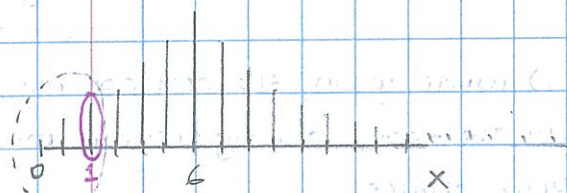
X = number of breakdowns

observed value $X = 1$ breakdown in 4 weeks

H_0 : $\lambda = 3$ per 2 weeks "no change" new copier is same as old
= 6 per 4 weeks

H_1 : $\lambda < 6$ per 4 weeks (more reliable)

NOTE: λ must be consistent with the "observed" time interval.



$$P(X \leq 1) = P(X=0, 1)$$

$$= e^{-6} (1 + 6)$$

$$= 0.017351$$

$$1.74\%$$

are these probs $< 5\%$. Since $1.74\% < 5\%$, we reject H_0 .
The new copier is more reliable.

HYPOTHESIS TEST FOR DISCRETE VARIABLES USING NORMAL APPROX.

N approx to Bn

$$np > 5$$

$$nq > 5$$

Example : multichoice exam

- 100 questions

- 4 possible answers

Michael gets 34 corrects

Test at 5% level whether Michael is guessing

$X = N^{\circ}$ he gets correct

If he is guessing then $X \sim (100, \frac{1}{4})$

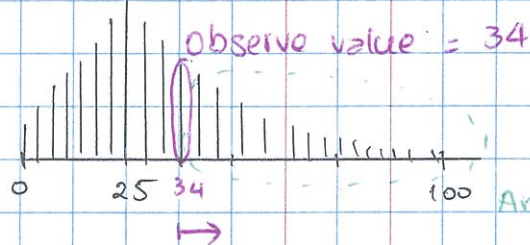
$$H_0 : p = \frac{1}{4}$$

$$H_1 : p > \frac{1}{4}$$

$$np = \frac{1}{4} \times 100 = 25$$

$$np = 25 > 5$$

$$nq = 75 > 5$$



$$P(X \geq 34) = P(Z \geq \frac{33.5 - 25}{\sqrt{100 \times \frac{1}{4} \times \frac{3}{4}}})$$

$$= P(Z \geq 1.9629..)$$

$$= 0.024824 = 0.0248$$

$$= 2.48\%$$

Since $2.48\% < 5\%$, we reject H_0 , so he is not guessing.

Exercise 7B

- 1 A jar contains a large number of coloured beads, some of which are red. A random sample of 80 of these beads is selected and 19 are found to be red. Test, at the 10% significance level, whether 30% of the beads in the jar are red.
- 2 A new cold relief drug is tested for effectiveness on 150 volunteers, and 124 of them found the drug beneficial. The manufacturers believe that more than 75% of people suffering from a cold will find the drug beneficial. Test the manufacturers' belief at the $2\frac{1}{2}\%$ significance level.
- 3 A selling price of \$22 000 has been proposed for a new model of car manufactured by a large company. The price will be adopted if more than 40% of potential customers are willing to pay that price. A sample of 50 potential customers were supplied with the car's specification and 29 agreed to pay the proposed price. Carry out a test, at the 1% significance level, to decide whether the company should adopt the price of \$22 000.
- 4 A parcel delivery service claims that at least 80% of their parcels are delivered within 48 hours of posting. A check on 200 parcels found that 152 were delivered within 48 hours of posting. Test the delivery service's claim at the 5% significance level.
- 5 In Zimbabwe, 8% of males are colour-blind. A random sample of 500 males was selected from a town in Zimbabwe and 53 men were found to be colour-blind. Find the p -value of the test of whether the town contains a different proportion of colour-blind males than in Zimbabwe as a whole.
What would be the conclusion of the test at the following significance levels:
(a) 5%, (b) 2%?
- 6 The drop-out rate of students enrolled at a certain university is reported to be 13.2%. The Dean of Students suspects that the drop-out rate for science students is greater than 13.2%, and she examines the records of a random sample of 95 of these students. The number of drop-outs was found to be 20. Test the Dean's suspicion at the 2% significance level.

① X - N° of red beads

$$n = 80 \quad p = 0.3 \quad np = 24 \quad P(X \leq 19) = P\left(Z \leq \frac{19.5 - 24}{\sqrt{80 \times 0.3 \times 0.7}}\right)$$

$$X \sim B(80, 0.3)$$

$$H_0: p = 0.3$$

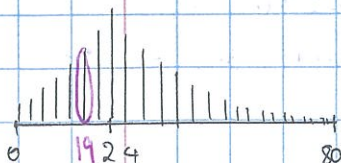
$$H_1: p \neq 0.3$$

$$= P(Z \leq -1.0978...)$$

$$= 0.13613...$$

$$= 13.61\%$$

since $13.61\% > 5\%$, we accept H_0 , so 30% of the beads are red.



observed value

2. X - N that found drug beneficial

$n = 150$ $p = 0.75$

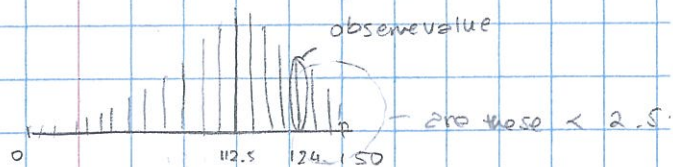
$np = 112.5 > 5$

$nq = 37.5 > 5$

$H_0: p = 0.75$

$H_1: p > 0.75$

observed value = 124



$$P(X \geq 124) = P\left(Z \geq \frac{123.5 - 112.5}{\sqrt{150 \times 0.75 \times 0.25}}\right)$$

$$= P(Z \geq 2.0741)$$

$$= 0.019031$$

$$= 1.9\%$$

1.9% < 2.5%, we reject H_0 , so more than 75% of people suffering from a cold will find the drug beneficial

3. X - agree to pay

$n = 50$ $p = 0.4$

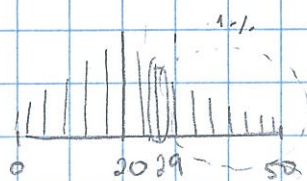
$H_0: p = 0.4$

$H_1: p > 0.4$

observed value = 29

$np = 20 > 5$

$nq = 30 > 5$



$$P(X \geq 29) = P\left(Z \geq \frac{28.5 - 20}{\sqrt{50 \times 0.4 \times 0.6}}\right)$$

$$= P(Z \geq 2.4537)$$

$$= 0.007069$$

$$= 0.71\%$$

0.71% < 1%, so we reject H_0

more than 40% of potential customers are willing to pay \$22,000, we should use the price of \$22,000 ✓

4.

$n = 200$ $p = 0.8$

X = delivered within 48 hrs

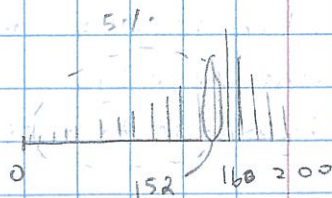
$H_0: p = 0.8$

$H_1: p < 0.8$

observe $X = 152$

$np = 200 \times 0.8 = 160 > 5$

$nq = 200 \times 0.2 = 40 > 5$



$$P(X \leq 152) = P\left(Z \leq \frac{152.5 - 160}{\sqrt{200 \times 0.8 \times 0.2}}\right)$$

$$= P(Z \leq -1.3258)$$

$$= 0.09244$$

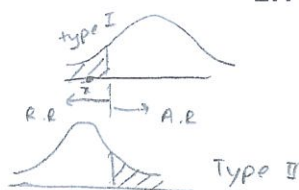
$$= 9.2\%$$

9.2% > 5%, we accept H_0

so, at least 80% of their parcels are delivered within 48 hours

ERRORS IN HYPOTHESIS TESTS

Errors in Hypothesis Tests – type I and II errors



		Decision	
		Accept H_0 (decide the coin is unbiased)	Reject H_0 (decide the coin is biased)
Reality	The null hypothesis, H_0 , is true.	Correct decision	H_0 wrongly rejected: Type I error
	The null hypothesis, H_0 , is false.	H_0 wrongly accepted: Type II error	Correct decision

In summary

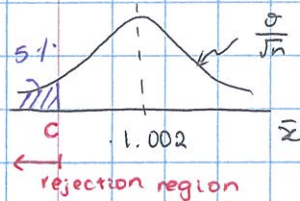
- A type I error occurs when the sample leads you to wrongly reject H_0 when it is in fact true.
- A type II error occurs when the sample leads you to wrongly accept H_0 when it is in fact false.

Example: A machine fills "1 L" bottles. When the machine is working correctly, the bottles are filled with water that is $N(1.002, 0.002^2)$ litres. The machine is tested by taking a sample of 9 bottles and calculating the sample mean. If the sample mean is below the certain value, we assume the machine is not working properly.

- set up null and alternative hypothesis to test if the machine is working properly
- find the rejection region for a 5% test
- Give the probability of making a type I error
- Find the probability of making a type II error, if the machine is not working properly and the population mean has fallen to 1.000 litres.

(a) $H_0: \mu = 1.002$
 $H_1: \mu < 1.002$

$\mu = 1.002$
 $\sigma = 0.002$
 $n = 9$
 $\bar{x} = ?$



(b) ~~Mark 645~~ Find c

$Z = -1.645$

$-1.645 = \frac{c - 1.002}{\frac{0.002}{\sqrt{9}}}$

Reject H_0 if $\bar{x} < 1.0009$

$c = 1.0009$

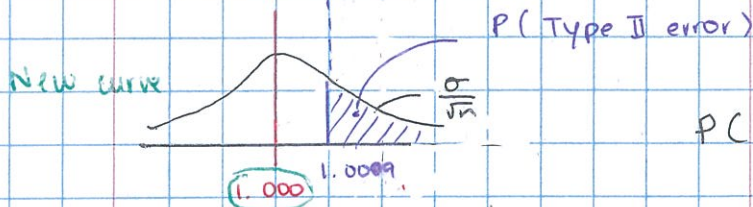
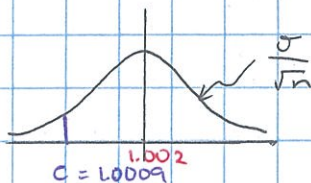
(c) NOTE: TYPE I error can only occur if we reject H_0 .

So, the probability of type I is equal to probability of the rejection region

$P(\text{Type I}) = P(\text{rejection region})$

Ans: 0.05

(d)



NOTE: A Type II error can only occur if we are in the acceptance region, but the probability is calculated using the new curve

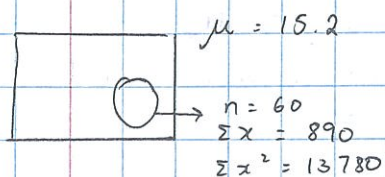
$$P(\text{Type II}) = P(\bar{x} > 1.0009) \\ = P\left(Z > \frac{1.0009 - 1.000}{\frac{0.002}{\sqrt{9}}}\right)$$

$$= P(Z > 1.355)$$

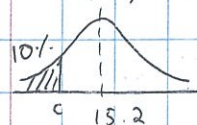
$$= 0.0877 \quad (3.5\%)$$

- 6 Photographers often need to take many photographs of families until they find a photograph which everyone in the family likes. The number of photographs taken until obtaining one which everybody likes has mean 15.2. A new photographer claims that she can obtain a photograph which everybody likes with fewer photographs taken. To test at the 10% level of significance whether this claim is justified, the numbers of photographs, x , taken by the new photographer with a random sample of 60 families are recorded. The results are summarised by $\Sigma x = 890$ and $\Sigma x^2 = 13780$.

- (i) Calculate unbiased estimates of the population mean and variance of the number of photographs taken by the new photographer. [3]
- (ii) State null and alternative hypotheses for the test, and state also the probability that the test results in a Type I error. Say what a Type I error means in the context of the question. [3]
- (iii) Carry out the test. [4]



(ii) $H_0: \mu = 15.2$
 $H_1: \mu < 15.2$



$P(\text{Type I}) = P(\text{rejection region})$
 $= 0.10$

A Type I error would occur when we reject H_0 . So, we conclude that she is quicker than at photographing, but actually she is not quicker.

(i) $\bar{x} = \frac{890}{60} = 14.833... = 14.8\bar{3}$

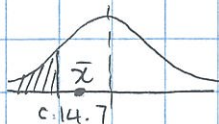
$s^2 = \frac{1}{59} \left(13780 - \frac{890^2}{60} \right)$
 $= 9.802259... = 9.80 \text{ (3sf)}$

(iii) The test calculate z
 $z = \frac{\bar{x} - \mu}{\frac{s}{\sqrt{n}}}$
 $z = -1.282$

$-1.282 = \frac{C - 15.2}{\sqrt{\frac{9.80}{60}}}$

$C = 14.6818...$

$C = 14.7 \text{ (3sf)}$



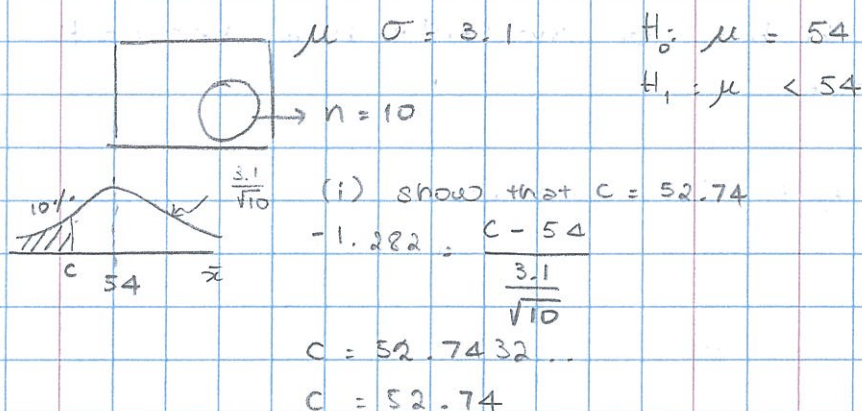
Since $14.8 > 14.7$ we accept H_0 .

Conclude that she photographs the same as everyone else.

- 5 The lectures in a mathematics department are scheduled to last 54 minutes, and the times of individual lectures may be assumed to have a normal distribution with mean μ minutes and standard deviation 3.1 minutes. One of the students commented that, on average, the lectures seemed too short. To investigate this, the times for a random sample of 10 lectures were used to test the null hypothesis $\mu = 54$ against the alternative hypothesis $\mu < 54$ at the 10% significance level.

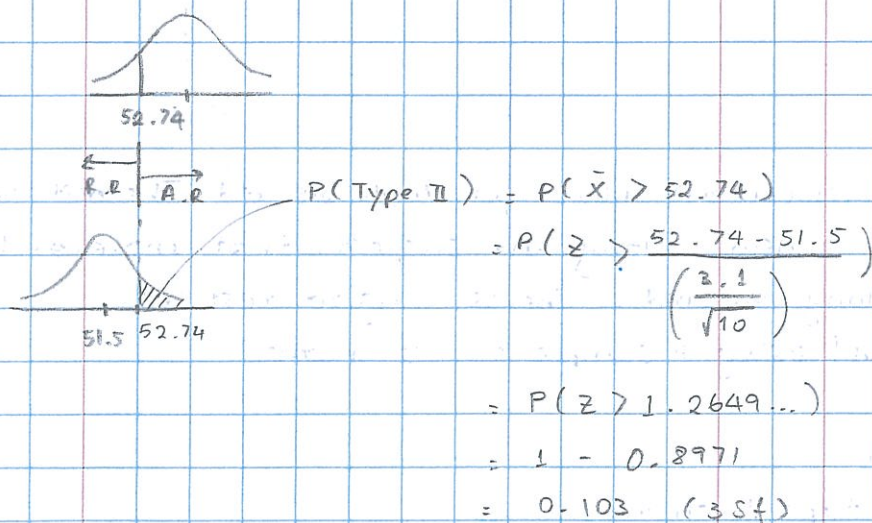
(i) Show that the null hypothesis is rejected in favour of the alternative hypothesis if $\bar{x} < 52.74$, where $\bar{x}$ minutes is the sample mean. [4]

(ii) Find the probability of a Type II error given that the actual mean length of lectures is 51.5 minutes. [4]



So, we will reject H_0 if $\bar{x} < 52.74$

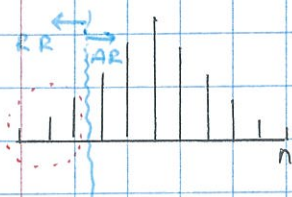
(ii)



ERRORS FOR DISCRETE VARIABLES TESTS

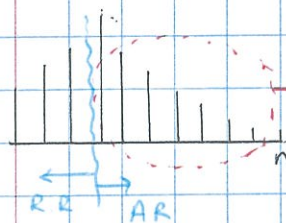
Type I: the "observed value" falls in the rejection region, so we reject H_0 but we shouldn't have because H_0 is actually true

Type II: the "observed value" falls in the acceptance region, so we accept H_0 but we shouldn't have because H_0 is actually false.



NOTE: this will not be a nice whole number

$$P(\text{Type I}) = P(\text{Rejection region})$$



New graph

$$P(\text{Type II})$$

It is known that 60% of certain species of moths are red. The rest are yellow. A biologist finds a new colony and suspects that more of the than 60% of the new colony are red. She is going to test this suspicion at the 5% level using a sample of 10 moths

1. State H_0 and H_1
2. find the rejection region
3. Find the probability of type I error
4. If actually 80% of the new colony are red, calculate the probability of type II error.

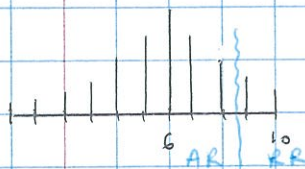
$$1. H_0: p = 0.6$$

$$H_1: p > 0.6$$

$$2. X = \text{N}^{\circ} \text{ of red moths}$$

$$X \sim B(10, 0.6)$$

$$np = 10 \times 0.6 = 6$$



$$P(X=10) = {}^{10}C_{10} (0.6)^{10} (0.4)^0 = 0.006047$$

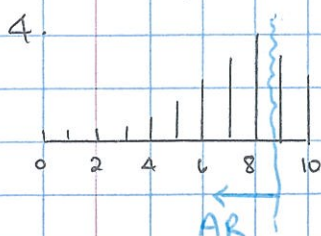
$$P(X=9) = {}^{10}C_9 (0.6)^9 (0.4) = 0.040311$$

$$P(X=8) = {}^{10}C_8 (0.6)^8 (0.4)^2 = 0.12093$$

$$P(X=7) = {}^{10}C_7 (0.6)^7 (0.4)^3 = 0.21499$$

$$P(X=6) = {}^{10}C_6$$

$$3. P(\text{Type I}) = 0.00605 + 0.0403 = 0.0463 \quad (3 \text{ sf})$$



New graph based
on $p=0.8$ $np=8$

we reject H_0 if $x \geq 9$
so we accept H_0 if $x < 9$

$$\begin{aligned} P(\text{Type II}) &= P(X=0, 1, 2, \dots, 8 \text{ using } p=0.8) \\ &= 1 - [{}^{10}C_9 (0.8)^9 (0.2) + {}^{10}C_{10} (0.8)^{10} (0.2)^0] \\ &= 1 - [0.2684 + 0.10737] \\ &= 1 - 0.3757 = 0.6242 \\ &= 0.624 \quad (3 \text{ sf}) \end{aligned}$$

Example ERRORS in a POISSON TEST

- 6 Pieces of metal discovered by people using metal detectors are found randomly in fields in a certain area at an average rate of 0.8 pieces per hectare. People using metal detectors in this area have a theory that ploughing the fields increases the average number of pieces of metal found per hectare. After ploughing, they tested this theory and found that a randomly chosen field of area 3 hectares yielded 5 pieces of metal.

(i) Carry out the test at the 10% level of significance. [6]

(ii) What would your conclusion have been if you had tested at the 5% level of significance? [1]

Jack decides that he will reject the null hypothesis that the average number is 0.8 pieces per hectare if he finds 4 or more pieces of metal in another ploughed field of area 3 hectares.

(iii) If the true mean after ploughing is 1.4 pieces per hectare, calculate the probability that Jack makes a Type II error. [3]

Observed value = 5 pieces in 3ha

$H_0: \lambda = 2.4$ pieces in 3ha $0.8/1\text{ha}$

$H_1: \lambda > 2.4$ pieces in 3ha



(i)

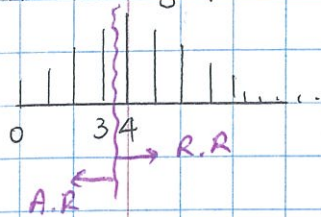
$$\begin{aligned} P(X \geq 5) &= 1 - P(X=0, 1, 2, 3, 4) \\ &= 1 - e^{-2.4} \left(1 + 2.4 + \frac{2.4^2}{2} + \frac{2.4^3}{3!} + \frac{2.4^4}{4!} \right) \\ &= 1 - 0.90413 \\ &= 0.095868 \\ &= 0.0959 \quad (3 \text{ sf}) \\ &= 9.59\% \end{aligned}$$

Since $9.59\% < 10\%$, we reject H_0 . So, Ploughing does increase number of pieces of metal.

(ii) Since $9.59\% > 5\%$, we accept H_0 .
Ploughing does not increase number of pieces of metal

$$1.4 \times 3 = 4.2$$

(iii) New graph



$$P(\text{Type II}) = P(\text{Acceptance region using } \lambda = 4.2)$$

$$= P(X = 0, 1, 2, 3)$$

$$= e^{-4.2} \left(1 + 4.2 + \frac{4.2^2}{2} + \frac{4.2^3}{3!} \right)$$

$$= 0.39540\dots$$

$$= 0.395 \quad (3 \text{ s.f.})$$